

## GE261004 DESIGN THINKING

Unit I – Introduction to Design Thinking · Presentation 6 of 6

# Case Studies of Successful Innovations

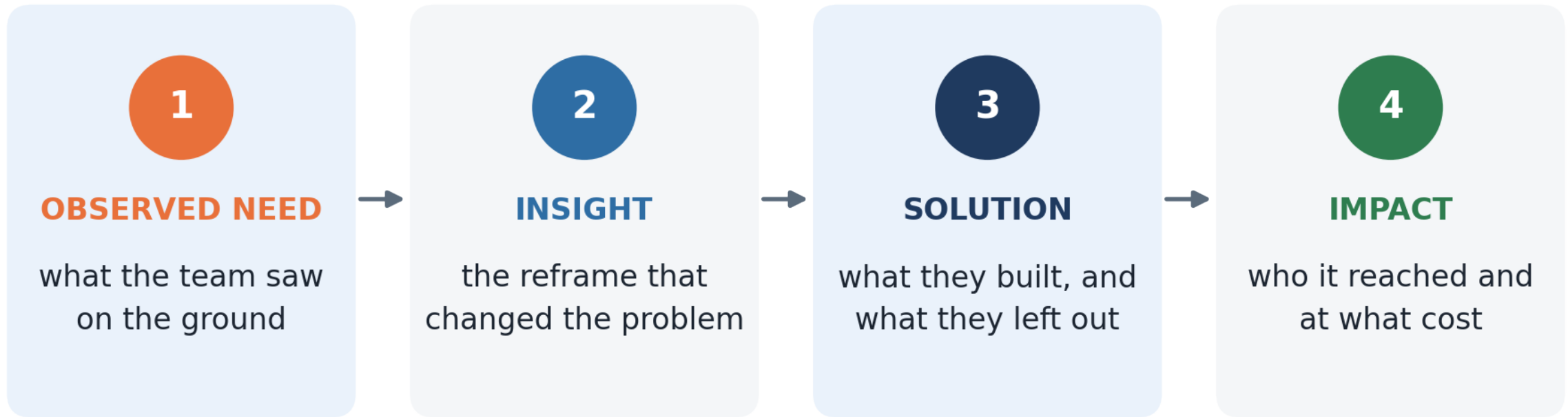
Embrace Infant Warmer · Aravind Eye Care · the Jaipur Foot · MittiCool · Tata Swach –  
and the reframe at the centre of each

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# In this presentation

- How to read a design case
- Embrace Infant Warmer
- Aravind Eye Care System
- The Jaipur Foot
- MittiCool
- Tata Swach
- The five cases side by side
- What they have in common

# How to read a design case



*the insight is where the design thinking happened*

Read every case in four moves – the interesting part is always the second.

# 1

## Health and care

Three cases where the reframe mattered more than the technology

# Case 1 – Embrace Infant Warmer

- A Stanford class was set a brief: build an incubator at one per cent of the usual \$20,000
- Linus Liang travelled to Nepal in 2007 and found the brief was wrong
- Most at-risk babies were born far from any hospital, where power is unreliable and staff are few
- A cheaper incubator in a city hospital would never have reached them

**KEY POINT** The team's real finding was not a design. It was that the problem they had been given was the wrong one.

# Embrace: the brief and the reframe

## The brief they were given

- Build a cheaper incubator
- Target: 1% of \$20,000
- For hospital use
- Mains electricity assumed

## The problem they found

- Keep newborns warm
- In villages, not hospitals
- Without reliable power
- Usable by the mother herself

**KEY POINT** Same babies, same goal – a completely different machine, because the problem statement changed.

# Embrace: what they built

- A small sleeping bag with a removable pouch of phase-change wax
- The pouch is warmed in hot water and holds about 37°C for roughly four hours
- No continuous electricity, no technician, and a mother can use it herself
- Material cost fell from about \$20,000 to the order of \$25

**KEY POINT** The technology is simple. The insight that made it the right technology was not.

## Case 2 – Aravind Eye Care System

- Founded in Madurai in 1976 by Dr Govindappa Venkataswamy with eleven beds
- The problem was not surgical technique – it was cost, distance and capacity
- Cross-subsidy: paying patients fund roughly half of all patients, who pay little or nothing
- An assembly-line theatre raised surgeon output far above the Indian average

**KEY POINT** Dr V studied how McDonald's achieved consistency at volume, and applied the same thinking to cataract surgery.

# Aravind: the second reframe

- Imported intraocular lenses cost hundreds of dollars and capped how many could be treated
- Rather than negotiate the price, Aravind set up Aurolab in 1992 and made them
- Lens cost fell to a few dollars; total surgery cost fell to around ten
- Today the system performs hundreds of thousands of surgeries a year

**KEY POINT** When a constraint blocks the mission, design thinking treats the constraint itself as the next problem.

## Case 3 – the Jaipur Foot

- Designed in 1968 by Dr P. K. Sethi, an orthopaedic surgeon, with Ram Chandra Sharma, a craftsman
- Western prosthetic feet assumed shoes, chairs and paved floors
- Indian users needed to walk barefoot, squat, sit cross-legged and work in wet fields
- Rubber and wood, fitted in a day, at a small fraction of the imported cost

**KEY POINT** The old foot was not badly engineered. It was engineered for a life its users did not live.

# 2

## Everyday products and public systems

Two Indian cases where the constraint became the design

## Case 4 – MittiCool

- Mansukhbhai Prajapati, a potter from Morbi district in Gujarat, worked in clay from 1989
- Households without electricity had no way at all to keep food from spoiling
- A two-chamber clay cabinet cools by evaporation from water held in its walls
- About 8°C below room temperature: vegetables keep four days, milk two

**KEY POINT** No electricity was treated as a design constraint to work within, not a problem to be solved first.

## Case 5 – Tata Swach

- Tata Chemicals' relief filters after the 2004 tsunami exposed what a household filter had to survive
- Launched in 2009: rice husk ash, a waste product, impregnated with nano-silver
- Gravity-fed – it needs neither electricity nor running water nor a plumber
- The cartridge stops the flow after about 3,000 litres so it cannot be used unsafely

**KEY POINT** Running cost was about ₹30 a month for a family of five – the number that decided whether it would be used at all.

# The five cases side by side

| Case               | The reframe              | What made it work                  |
|--------------------|--------------------------|------------------------------------|
| <b>Embrace</b>     | Villages, not hospitals  | Wax pouch, no mains power          |
| <b>Aravind</b>     | Volume, not technique    | Cross-subsidy and own lens factory |
| <b>Jaipur Foot</b> | Barefoot life, not shoes | Rubber and wood, fitted in a day   |
| <b>MittiCool</b>   | No electricity at all    | Evaporative cooling in clay        |
| <b>Tata Swach</b>  | No power, no plumbing    | Rice husk ash and nano-silver      |

# What all five have in common

- Somebody went and looked, and came back with a different problem
- The binding constraint – cost, power, distance – became the design brief
- The technology is modest; the framing is what was clever
- Each was designed for how people actually live, not how they ought to live
- All five reached scale, which is what separates innovation from invention

**KEY POINT** In every case, the decisive work happened before any engineering began.

# In class: read a case in four moves

- Each team takes one of the five cases, or brings another
- Write the observed need, the insight, the solution and the impact – one line each
- Then state what the brief would have been without the field visit
- Present in three minutes; the class judges whether the reframe is real

**ACTIVITY** Twenty minutes in teams. Every team presents; keep strictly to three minutes.

## THINK ABOUT IT

**Name a problem on this campus that you think is being solved from the wrong problem statement.**

- What is the stated problem, and who stated it?
- Who lives with it daily, and have they been asked?
- What would you find out by watching for one hour?
- What would the brief become if you did?

# Summary

- Embrace: the brief said incubator; the field said portable warmer
- Aravind: volume and cost, not surgical technique – then its own lens factory
- Jaipur Foot: designed for squatting, wet fields and bare feet
- MittiCool: no electricity treated as the design constraint
- Tata Swach: a waste material turned into an affordable filter
- In all five, the reframe mattered more than the technology

# Next: Unit II – Empathy and Problem Definition

- Understanding users and stakeholders
- Empathy mapping and observation techniques
- Interviews and need finding
- Framing problem statements, POV and reframing